

Chapter 5 Determinants

- **5.1 Properties of Determinants:**

- Calculate the determinant of a matrix using properties of determinants.
 - ex.** HW 1ab.
 - ex.** Let $\det(A) = \sqrt{2}$. Does A^{-1} exist? If so find $\det(A^{-1})$.
 - ex.** Let R be the reduced row echelon form of A . How does $\det A$ relate to $\det R$?
 - ex.** Let $A \in \mathcal{M}_{2 \times 2}$ with a determinant of 3 and $B = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$, find $\det AB$.
- For $n \times n$ matrices:
 - * See Proposition 1.7.
 - * A is an upper (or lower) triangle matrix $\Rightarrow \det A$ is equal to the product of A 's diagonal entries.
 - * A is nonsingular $\iff \det A \neq 0$.
 - * $\det(AB) = \det A \det B$.
 - * A is a nonsingular $\Rightarrow \det(A^{-1}) = \frac{1}{\det A}$.

- **5.2 Cofactors**

- Calculate the determinant of a matrix using cofactors.
- Calculate the x_i coordinate in a solution vector using Cramer's rule.

Chapter 6 Eigenvalues and Eigenvectors

- **6.1: The Characteristic Polynomial**

- Calculate the characteristic polynomial of A , $p_A(t)$.
- Find the eigenvalues and eigenvectors of A .
- Know the definition of *eigenvalue* and *eigenvector*.

- **6.2: Diagonalization**

- Know the definition of *similar matrices*: A and B are similar if there exists an invertible matrix P such that $A = P^{-1}BP$
- Prove: If A and B are similar matrices then $\det(A) = \det(B)$. (Hint: use the last two properties under 5.1 above)
- Know the definition of *diagonalizable*.